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Jaeyun (Jay) Ree

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LinkedIn,GitHub,Website

Final-year Software Engineering (Honours) student at Monash University, graduating Nov 2026, with research interests in quantum and classical algorithms for combinatorial optimisation. As a Research Assistant under Dr Mohammed Eunus Ali, I designed and benchmarked matching and assignment algorithms for autonomous-vehicle coordination, resulting in a manuscript now being revised for resubmission following peer review. I also completed a Human Research Ethics-approved Honours thesis on human-centred adaptive assessment, and bring software engineering experience as co-founder and CTO of AIVA. I am a recipient of the 2023 and 2024 Monash Engineering Dean's Honours.

Publications & Research Output

Under review

- J. Ree, A. B. M. Russel, and J. Grundy, "An Empirical Study of Context-Aware LLM-Adaptive Mental Health Questionnaires," submitted to the Australian Conference on Human-Computer Interaction (OzCHI 2026). *Under review.*

In revision

- J. Ree, M. E. Ali, and M. A. Cheema, "Dynamic Platoon Formation: Spatio-Temporal Matching for Towing-Based Cooperative Autonomous Mobility." Manuscript in revision for resubmission following peer review; extended evaluation on SUMO-generated road-network data in progress.

Research Experience

Research Assistant, Monash University, Clayton Victoria Nov 2025 -
Project: Dynamic Platoon Formation of Multi-Type Autonomous Vehicles for Sustainable Urban Mobility

- Contributed to modelling the coordination task as a constrained optimisation problem with multi-segment matching and point-wise capacity constraints, under Dr Mohammed Eunus Ali (with Prof. Aamir Cheema as co-author).
- Designed and implemented two matching algorithms, a Greedy Maximum-Weight Matching baseline and an Iterative Linear Assignment method using the Jonker-Volgenant solver, and evaluated them on synthetic scenarios with up to 400 passive and 80 active vehicles.
- Built an end-to-end experimental framework in **Python** (NumPy, SciPy, Pandas) with parameter-sweep experiments and comparative benchmarking; revised and resubmitted the work across two conference formats in response to peer review.

Final Year Project Researcher (Honours Thesis), Monash University Jul 2025 -
Human-Centred Adaptive Assessment (FIT4701 / FIT4702)

- Conducted a human-centred study comparing rule-based and LLM-based adaptive mental-health questionnaires, supervised by Dr ABM Russel with co-supervision from Prof. John Grundy (ARC Laureate Fellow) and industry mentorship from Mark Foley (Future Wellness Group).
- Designed and ran a MUHREC-approved within-subjects study (Project ID 51664) with 40 participants, grounded in the COSMIN content-validity framework, and co-developed "Navigator", a Flutter/Supabase prototype implementing both adaptation methods.
- Led the statistical analysis (paired-sample t-tests, effect sizes) and authored the final report; the LLM-based method scored significantly higher on all three dimensions (Clarity, Relevance, Quality). The work has been written up and submitted to OzCHI 2026.

Professional Experience

Teaching Associate, Monash University, Melbourne Victoria **Mar 2026 - Jun 2026**

- Delivered weekly applied sessions for *FIT1045: Introduction to Programming*, mentoring students on C++ fundamentals and the SplashKit library, and assessing portfolio submissions against structured rubrics.

Chief Technology Officer (CTO), AIVA, Melbourne **Dec 2025 -**

- Co-founded and led the end-to-end architecture of a full-stack AI executive-assistant platform (Node.js/Express and Supabase/PostgreSQL backend, Flutter iOS client, and an autonomous Google Gemini agent), directing technical strategy and code review across the team.

Full Stack Developer Intern, Job Search Works, Melbourne **May 2025 - Nov 2025**

- Built core features for an AI-powered job-application SaaS (resume-job matching, automated workflows) using **React**, **TypeScript**, and **Firebase**; set up CI/CD with **GitHub Actions** and Husky.

IT Analyst, Hardware Co-Ordination Intern, Coles Group **Jul 2024 - Dec 2024**

- Coordinated hardware lifecycle management across Coles divisions; automated a manual reporting process with VBA (4 hours to 30 minutes) and contributed to over $90,000 in documented operational savings through improved vendor and lifecycle planning.

Education

Monash University **July 2020 - Present [Expected Graduation: Nov 2026]**

- Bachelor of Engineering (Honours) in Software Engineering; GPA 3.759 / 4.000, WAM 82.571 (High Distinction).
- Relevant coursework in algorithms and data structures, discrete mathematics, databases, and operating systems. Proficient in **Python** (NumPy, SciPy, Pandas), **Java**, and **TypeScript/JavaScript**; experienced with Git, Linux, and SQL.

Monash College (Foundation Year) **July 2019 - June 2020**
GPA 4.0/4.0 (High Distinction)

Awards & Honours

2024 Monash Engineering Dean's Honours (Jun 2025)
2023 Monash Engineering Dean's Honours (May 2024)
Summer Vacation Research Scholarship, Faculty of IT (2025)
Industry-Based Learning Placement Scholarship (Jul 2024)
Academic Commendation for Highest Grade in FIT2081: Mobile Application Development (May 2024)
Engineering International Undergraduate Excellence Scholarship (Jun 2020)

Certifications

AWS Certified Cloud Practitioner (Mar 2025)

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